

Consultație ec. diferențiale

↳ rezolvări fără formule

(1) Soluția ec. diferențiale

$$\boxed{(1+x^2)y' = -xy \;,\; y(0)=2}$$

$\frac{dy}{dx} = y' = -\frac{xy}{1+x^2}$ → ec. diferențială de ord. I ⇒

↳ funcție ↳ variabilă

formă normală

$$\frac{dy}{y} = \frac{-x}{1+x^2}dx \Rightarrow \int \frac{dy}{y} = -\frac{1}{2}\int \frac{2x}{1+x^2}dx \Rightarrow$$

$$\ln|y| = -\frac{1}{2}\ln(1+x^2) + C \overset{\ln e^C}{\Rightarrow} \ln\frac{e^C}{\sqrt{1+x^2}} = \ln|y| \Rightarrow$$

$$|y| = \frac{e^C}{\sqrt{1+x^2}} \Rightarrow y = \pm\frac{e^C}{\sqrt{1+x^2}}, \; c \text{ constantă}$$

$$\Rightarrow y(x) = \frac{k}{\sqrt{1+x^2}} \Rightarrow 2 = \frac{k}{\sqrt{1+0^2}} \Rightarrow k=2 \Rightarrow$$

$y(0)=2$

$$\boxed{y(x) = \frac{2}{\sqrt{1+x^2}}}$$

(2) $xy' + (1+x)y = e^{-x}\sin 2x$, $y\left(\frac{\pi}{4}\right) = 0$ ← condiție inițială

$$y' = \frac{e^{-x}\sin 2x - (1+x)y}{x} = -\frac{1+x}{x}\,y + \frac{e^{-x}\sin 2x}{x}$$

$$y' = a(x)\cdot y + b(x)$$

↑ partea omogenă

I Rezolvăm ecuația omogenă atașată

$$y' = -\frac{1+x}{x}\cdot y \Rightarrow \frac{dy}{y} = \frac{-1+x}{x}\,dx \Rightarrow \ln|y| = -\int \frac{1+x}{x}\,dx$$

$$= -\ln|x| + x + C = \ln\frac{e^C \cdot e^{-x}}{|x|} \Rightarrow$$

$$|y| = \frac{e^C e^{-x}}{|x|} \Rightarrow y = \pm e^C \frac{e^{-x}}{x} \Rightarrow \boxed{y = k\cdot\frac{e^{-x}}{x}}$$

↳ poate fi orice număr real

II Metoda variației constantei

„S-a emancipat constanta"



Căutăm $y = \frac{k(x)\cdot e^{-x}}{x} \Rightarrow$

$$y' = \frac{[k'(x)\cdot e^{-x} + k(x)(-e^{-x})]\,x - (k(x)\cdot e^{-x})\cdot 1}{x^2}$$

$$= -\frac{1+x}{x}\cdot\underbrace{\frac{k(x)\cdot e^{-x}}{x}}_{y} + \frac{e^{-x}\sin 2x}{x} \Leftrightarrow$$

$$\frac{k'(x)\cdot e^{-x}}{x} - \frac{k(x)\cdot e^{-x}}{x} - \frac{k(x)\,e^{-x}}{x^2} = -\frac{k(x)\cdot e^{-x}}{x^2}\cdot(1+x)$$

$$+ \frac{e^{-x}\sin 2x}{x} \Leftrightarrow k'(x) = \sin 2x \Leftrightarrow k(x) = \int \sin 2x\,dx$$

$$\Leftrightarrow k(x) = -\frac{\cos 2x}{2} + C_1 \Rightarrow$$

$$y = \frac{e^{-x}}{x} \cdot \left(-\frac{\cos 2x}{2} + C_1\right)$$

Condiție

$y\left(\frac{\pi}{4}\right) = 0 \Rightarrow$

$$\frac{e^{-\pi/4}}{\pi} \cdot 4 \cdot C_1 = 0 \Rightarrow \boxed{C_1 = 0} \Rightarrow \boxed{y = \frac{e^{-x}}{x} \cdot -\frac{\cos 2x}{2}}$$

(3.) $(x^2+y^2)\,y' = xy$, $y(2) = 1$ → ecuație omogenă

$$y' = \frac{xy}{x^2+y^2} \quad \backslash x^2 \Rightarrow y' = \frac{\frac{y}{x}}{1+\left(\frac{y}{x}\right)^2} = f\left(\frac{y}{x}\right)$$

$$\frac{y}{x} = u(x) \Rightarrow y' = x \cdot u \Rightarrow u + x \cdot u' = \frac{u}{1+u^2} \Rightarrow$$

$$u' = \frac{\frac{u}{1+u^2} - u}{x} = -\frac{u^3}{x(1+u^2)} \quad \text{(EVS)}$$

$$\frac{1+u^2}{u^3}\,du = -\frac{dx}{x} \quad - - - -$$

Citat Negrescu:
„Să convertim formulițele în demonstrații"

(4) $y' = -\frac{y}{x} - y^2 \ln x$, $y(1) = 1$ → Bernoulli

$$\boxed{y' = a(x) \cdot y + b(x) \cdot y^\alpha}$$

$$y' = -\frac{1}{x} \cdot y + (-\ln x) \cdot y^2$$

$u = y^{1-\alpha}$ → substituție

$u = y^{1-2} = y^{-1} \Rightarrow y = \frac{1}{u}$

se reduce

$$-\frac{u'}{u^2} = \frac{-1}{ux} - \frac{\ln x}{u^2} \quad /\cdot(-u^2) \Rightarrow$$

$$\boxed{u' = \frac{u}{x} + \ln x} \rightarrow \text{ecuație liniară}$$

I $u' = \frac{u}{x} \Rightarrow \frac{du}{dx} = \frac{u}{x} \Rightarrow \frac{du}{u} = \frac{dx}{x} \Rightarrow \ln|u| = \ln|x| + C \Rightarrow$

$|u| = e^C \cdot |x| \Rightarrow u = \pm e^C \cdot x \Rightarrow u = k \cdot x$, k constantă

II Metoda variației constantelor

$u = k(x) \cdot x \Rightarrow k'(x)\,x + k(x) = k(x) + \ln x \Rightarrow$

$k'(x) = \frac{\ln x}{x} \Rightarrow k(x) = \int \frac{\ln x}{x} dx \Rightarrow k(x) = \frac{\ln^2 x}{2} + C \Rightarrow$

$u = x\left(\frac{\ln^2 x}{2} + C\right) \Rightarrow y = \frac{1}{u} \Rightarrow y = \frac{1}{x\left(\frac{\ln^2 x}{2} + C_1\right)}$

$y(1) = 1 \Rightarrow \frac{1}{C_1} = 1 \Rightarrow \boxed{C_1 = 1}$



(5) $y' = y^2 - \frac{y}{x} - \frac{1}{x^2}$

$x > 0$, știind că admite soluția particulară $y_p = \frac{1}{x}$ și că $y(1) = 2$

Soluție

$y = y_p + u = \left(\frac{1}{x} + u\right)$

Ecuație Riccati $\longrightarrow y' = a(x)y^2 + b(x)y + c(x)$

$-\frac{1}{x^2} + u' = \frac{1}{x^2} + \frac{2u}{x} + u^2 - \frac{\frac{1}{x} + u}{x} - \frac{1}{x^2} \Leftrightarrow$

$u' = \frac{2u}{x} + u^2 - \frac{u}{x} \Leftrightarrow \left(u' = u^2 + \frac{u}{x}\right) \rightarrow$ Ecuație Bernoulli

$z = u^{1-2} = \left(u^{-1}\right) \rightarrow$ ecuația se reduce la o liniară și se rezolvă liniara

(6) $\underbrace{(xy^2 - \sin x \cos x)}_{P} dx + \underbrace{y(x^2 - 1)}_{Q} dy = 0$

$y(0) = 0$

ω 1-formă diferențială

Ecuație exactă

$\left.\begin{aligned} \frac{\partial P}{\partial y} &= 2xy \\ \frac{\partial Q}{\partial x} &= 2xy \end{aligned}\right\} \Rightarrow \frac{\partial P}{\partial y} = \frac{\partial Q}{\partial x} \Rightarrow$ (ω închisă)

$\mathbb{R}^2$ domeniu stelat $\Longrightarrow$ T. Poincaré

ω exactă $\Rightarrow \exists F \in C^2$ a.î. $\omega = dF \Rightarrow$

$\left\{\begin{aligned} \frac{\partial F}{\partial x} &= P = xy^2 - \frac{\sin x \cos x}{2} \\ \frac{\partial F}{\partial y} &= Q = yx^2 - y \end{aligned}\right.$

$\Rightarrow F = \int (yx^2 - y)\,dy = x^2\frac{y^2}{2} - \frac{y^2}{2} + C(x)$

$\Rightarrow$ ~~xy^2~~ $- C'(x) =$ ~~xy^2~~ $- \frac{\sin 2x}{2} \Rightarrow$

$C'(x) = \frac{-\sin 2x}{2} \Rightarrow C(x) = \frac{\cos 2x}{4} + C_1$ → constantă

$\Rightarrow F = \frac{x^2y^2}{2} - \frac{x^2}{2} + \frac{\cos 2x}{4} + C_1$

$\omega = 0 \Rightarrow \frac{\partial F}{\partial x}dx + \frac{\partial F}{\partial y}dy = 0 \;|\cdot dx \Rightarrow \frac{\partial F}{\partial x} + \frac{\partial F}{\partial y}\cdot\frac{dy}{dx} = 0$

F

x y → x

$\Rightarrow \frac{d}{dx}(F(x, y(x))) = 0 \rightarrow F$ fct const

— — —

(7) $\underbrace{(1 - x^2y)}_{P}\,dx + \underbrace{(x^2y - x^3)}_{Q}\,dy = 0 \;|\; \boxed{x > 0}$

$\begin{cases} \frac{\partial P}{\partial y} = -x^2 \\ \frac{\partial Q}{\partial x} = 2xy - 3x^2 \end{cases}$

$P\,dx + Q\,dy = 0 \;|\cdot h \Rightarrow$ ec. exactă

$Ph\,dx + Qh\,dy = 0$

$P_1 \quad Q_1$

$\frac{\partial P_1}{\partial y} = \frac{\partial Q_1}{\partial x}$

$\frac{\partial P_1}{\partial y} = \frac{\partial P_2}{\partial x} =$

$\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} = 2xy - 2x^2 = 2x(y - x)$

$h' = f \cdot h \Rightarrow h' = \frac{2}{x}h \Rightarrow \frac{dh}{dx} = \frac{2h}{x}$, $\ln|h| = 2\ln|x| - c$

$\Rightarrow |h| = \frac{k}{x^2} \Rightarrow \boxed{h = \frac{1}{x^2}} \Rightarrow$

$\underbrace{\left(\frac{1}{x^2} - y\right)}_{P_1}dx + \underbrace{(y - x)}_{Q_1}dy = 0 \rightarrow$ ecuație exactă

Ecuații liniare cu coef. constanți

$$y'' - 4y' + 4y = (x+1)e^x + 2e^{2x} - \frac{e^{2x}}{(x+1)^3}$$

I $y'' - 4y' + 4y = 0 \longrightarrow \lambda^2 - 4\lambda + 4 = 0$

$\lambda_{1,2} = 2 \longrightarrow e^{2x}, xe^{2x}$

$y_0(x) = C_1 \cdot e^{2x} + C_2 x e^{2x}$

II $y'' - 4y' + 4y = (x+1)e^x = e^{ax}(P(x)\cos bx + Q(x)\sin bx)$

$\begin{cases} a = 1 \\ b = 0 \end{cases}$ $P(x) = x+1$

$y_{p1} = e^x((cx+d)\cos 0x) = e^x(cx+d) \longrightarrow$ soluție particulară

$y_{p1}'' = e^x(cx + 2c + d)$ $y_{p1}' = e^x(cx + c + d)$

$\Rightarrow e^x(cx + 2c + d) - 4e^x(cx + d + c) + 4e^x(cx + d) = (x+1)e^x$

$\Rightarrow \begin{cases} c = 1 \\ d = 3 \end{cases} \Rightarrow \boxed{y_{p1} = e^x(x+3)}$

III $y'' - 4y' + 4y = 2e^{2x} = e^{ax}(P\cos bx + Q\sin bx)$

$a = 2$, $b = 0$, $P = 2$, Q nu contează

$y_{p2} = e^{2x} \cdot k \rightarrow 4ke^{2x} - 4 \cdot 2ke^{2x} + 4ke^{2x} = 2e^{2x}$

$4ke^{2x} - 4 \cdot 2ke^{2x} + 4ke^{2x} = 0$

↓ soluția ecuației omogene

Dacă $a+bi$ este rădăcină a polinomului caracteristic de multiplicitate m se înmulțește cvasipolinomul cu $x^m \Rightarrow$ caut particulara de forma $\boxed{x^2 e^{2x}}$

$y_p'' = c(2e^{2x} + 4xe^{2x} + 4xe^{2x} + 4x^2e^{2x})$

$2ce^{2x} + 8cxe^{2x} + 4cx^2e^{2x} - 8cxe^{2x} - 8cx^2e^{2x} + 4cx^2e^{2x} =$

$= 2e^{2x} \Rightarrow 2C = 2 \Rightarrow C = 1 \Rightarrow \boxed{y_{p_2} = x^2 e^{2x}}$

IV $y'' - 4y' + 4y = \dfrac{-e^{2x}}{(x+1)^3}$

Aplicăm metoda variației constantelor

$$\begin{cases} y_{p_3} = C_1(x)\, e^{2x} + C_2(x) \cdot x e^{2x} \\ y_{p_3} = C_1(x) \cdot y_1(x) + C_2(x) \cdot y_2(x) \end{cases}$$

$$\begin{cases} C_1' y_1 + C_2' y_2 = 0 \\ C_1' y_1' + C_2' y_2' = f(x) \end{cases}$$

$$\begin{cases} C_1' \cdot e^{2x} + C_2' \cdot x e^{2x} = 0 \\ C_1' \cdot 2e^{2x} + C_2' (e^{2x} + 2x e^{2x}) = \dfrac{-e^{2x}}{(x+1)^3} \end{cases} \Rightarrow$$

$$C_2' \cdot e^{2x} = \frac{-e^{2x}}{(x+1)^3} \iff C_2' = \frac{-1}{(x+1)^3} \Rightarrow$$

$\boxed{C_2 = \dfrac{1}{2(x+1)^2} + k_2} \rightarrow$ alegem $k_2 = 0$ (soluție particulară)

$$C_1' \cdot e^{2x} - \frac{1}{(x+1)^3} \cdot x e^{2x} = 0 \Rightarrow C_1' = \frac{x}{(x+1)^3} \Rightarrow$$

$$\Rightarrow \boxed{C_1 = -\frac{1}{x+1} + \frac{1}{2(x+1)^2}} \Rightarrow$$

$$\boxed{y_{p_3} = \left(\frac{-1}{x+1} + \frac{1}{2(x+1)^2}\right) \cdot e^{2x} + \frac{1}{2(x+1)^2} \cdot x e^{2x}}$$

$$\Rightarrow \boxed{y(x) = y_0(x) + y_{p_1}(x) + y_{p_2}(x) + y_{p_3}(x)}$$

(9.) $x^2 y'' - x y' + y = \sin(\ln x), \; x > 0$

$x = e^t$

$y(x) = y(e^t) = z(t)$

$\downarrow \; y'(e^t) \cdot e^t = z'(t) \Rightarrow y'(e^t) = e^{-t} \cdot z'(t)$

$y''(e^t)\cdot e^t = -e^{-t}\cdot z'(t) + e^{-t}\, z''(t)$

$y''(e^t) = e^{-2t}(z''(t) - z'(t))$

$e^{2t}\cdot e^{-2t}(z''-z') - e^{t}\, e^{t} z' + z = \sin t$

$z'' - z' - z' + z = \sin t \Rightarrow \boxed{z'' - 2z' + z = \sin t}$

Sisteme

(10.) $\begin{cases} x_1' = -2x_1 + x_2 + 2e^{-t} \\ x_2' = x_1 - 2x_2 + 1 \end{cases}$

$\boxed{\begin{array}{l} x_1(0) = 1 \\ x_2(0) = 0 \end{array}}$

Metoda substituției

$x_2 = x_1' + 2x_1 - 2e^{-t}$

$x_1'' + 2x_1' + e^{-t}\cdot 2 = x_1 - 2x_1 - 4x_1 + 4e^{-t} + 1$

$\boxed{x_1'' + 4x_1' + 3x_1 = 2e^{-t} + 1}$ ---

(11.) Metoda matricii exponențiale

$\begin{cases} x_1' = 5x_1 - 3x_2 + e^t \\ x_2' = -6x_1 + 2x_2 + 1 \end{cases}$

$\underbrace{\begin{pmatrix} x_1 \\ x_2 \end{pmatrix}'}_{X'} = \underbrace{\begin{pmatrix} 5 & -3 \\ -6 & 2 \end{pmatrix}}_{A}\underbrace{\begin{pmatrix} x_1 \\ x_2 \end{pmatrix}}_{X} + \underbrace{\begin{pmatrix} e^t \\ 1 \end{pmatrix}}_{b}$

$\Rightarrow \boxed{X' = AX + b}$

$x'(s) = A\cdot x(s) + b(s) \Rightarrow \cdot e^{-sA} \Rightarrow x'(s) - Ax(s) = b(s) \Rightarrow$

$e^{-sA}\cdot x'(s) - A\cdot e^{-sA}\cdot x(s) = e^{-sA}\, b(s)$

$\frac{d}{ds}\left(e^{-sA}\cdot x(s)\right) = e^{-sA}\, b(s)$

$$e^{-sA}\cdot x(s)\Big|_0^t = \int_0^t e^{-sA}\,b(s)\,ds$$

$$e^{-tA}\cdot x(t) - x(0) = \int_0^t \dots \Rightarrow \boxed{x(t) = e^{tA}\cdot x(0) + e^{tA}\int_0^t e^{-sA}\cdot b(s)\,ds}$$

$$A = \begin{pmatrix} 5 & -3 \\ -6 & 2 \end{pmatrix} \quad \det(A-\lambda I_2) = \begin{vmatrix} 5-\lambda & -3 \\ -6 & 2-\lambda \end{vmatrix} = \boxed{\lambda^2 - 7\lambda - 8}$$

$$= (\lambda+1)(\lambda-8) \Rightarrow \boxed{\lambda_1 = -1},\ \boxed{\lambda_2 = 8}$$

$\lambda = -1 \Rightarrow Av = \lambda v \Rightarrow \begin{cases} 5v_1 - 3v_2 = -v_1 \\ -6v_1 + 2v_2 = -v_2 \end{cases} \Rightarrow v_2 = 2v_1 \Rightarrow$

$$\boxed{V_{\lambda_1} = \mathrm{Sp}\left\{\begin{pmatrix} 1 \\ 2 \end{pmatrix}\right\}}$$

$\lambda = 8 \Rightarrow Av = 8v \Rightarrow \begin{cases} 5v_1 - 3v_2 = 8v_1 \\ -6v_1 + 2v_2 = 8v_2 \end{cases} \Rightarrow v_2 = -v_1 \Rightarrow$

$$\boxed{V_{\lambda_2} = \mathrm{Sp}\left\{\begin{pmatrix} 1 \\ -1 \end{pmatrix}\right\}} \Rightarrow$$

$$D = \begin{pmatrix} -1 & 0 \\ 0 & 8 \end{pmatrix} \rightarrow S = \begin{pmatrix} 1 & 1 \\ 2 & -1 \end{pmatrix}$$

$$S^{-1} = ? = -\frac{1}{3}\begin{pmatrix} -1 & -1 \\ -2 & 1 \end{pmatrix}$$

$$\boxed{A = SDS^{-1}} \Rightarrow e^{tA} = \sum_{n=0}^{\infty} \frac{(tA)^n}{n!} = S\cdot e^{tD}\cdot S^{-1}$$

$$e^{tA} = \begin{pmatrix} 1 & 1 \\ 2 & -1 \end{pmatrix}\cdot\begin{pmatrix} e^{-t} & 0 \\ 0 & e^{8t} \end{pmatrix}\begin{pmatrix} -1 & -1 \\ -2 & 1 \end{pmatrix} = -\frac{1}{3}\begin{pmatrix} e^{-t} & e^{8t} \\ 2e^{-t} & -e^{8t} \end{pmatrix}\begin{pmatrix} -1 & -1 \\ -2 & 1 \end{pmatrix}$$

$$= -\frac{1}{3}\begin{pmatrix} -e^{-t} - 2e^{8t} & -e^{-t} + e^{8t} \\ -2e^{-t} + 2e^{8t} & -2e^{-t} - e^{8t} \end{pmatrix}$$

condiții inițiale

$$x(t) = -\frac{1}{3}\begin{pmatrix} -e^{-t} - 2e^{8t} & -e^{-t} + e^{8t} \\ -2e^{-t} + 2e^{8t} & -2e^{-t} - e^{8t} \end{pmatrix}\begin{pmatrix} 6 \\ 1 \end{pmatrix} +$$

$$\left(-\frac{1}{3}\right)\begin{pmatrix} -e^{-t} - 2e^{8t} & -e^{-t} + e^{8t} \\ -2e^{-t} + 2e^{8t} & -2e^{-t} - e^{8t} \end{pmatrix}\cdot\int_0^t \left(-\frac{1}{3}\right)\cdot\begin{pmatrix} -e^{s} - 2e^{-8s} & -e^{s} + e^{-8s} \\ -2e^{s} + 2e^{-8s} & -2e^{s} - e^{-8s} \end{pmatrix}\cdot$$

$$\begin{pmatrix} e^{s} \\ 1 \end{pmatrix} ds = \dots$$